

Privacy-Preserving Platoon Control—Constrained Cooperative-Tracking Control via Time-Varying Heterogeneous Directed Networks

Lingying Huang^{ID}, Rong Su^{ID}, *Senior Member, IEEE*, Maode Ma^{ID}, *Senior Member, IEEE*, Yun Lu^{ID}, *Member, IEEE*, and Peihu Duan

Abstract—Distributed cooperative tracking control has emerged as a pivotal research focus in multi-agent systems, particularly for platoon control applications where its decentralized architecture offers significant advantages over centralized approaches. However, the direct exchange of sensitive data between agents raises critical privacy risks, hindering its broader adoption across safety-critical applications. This paper presents a privacy-preserving cooperative tracking framework that rigorously maintains bounded coupling errors, which is a crucial requirement for collision avoidance in vehicular platoons. Departing from conventional methods that compromise privacy through explicit state sharing for error mitigation, our proposed algorithm achieves dual objectives: maintaining prescribed error constraints while preserving agent state confidentiality in directed communication networks with time-varying interaction weights. We establish sufficient conditions for achieving cooperative-tracking consensus with predefined error constraints and characterize the quantitative relationship between the asymptotic convergence rate and control gain parameters. Furthermore, we analyse the privacy-preserving performance against internal and external adversaries, demonstrating that the probability of an adversary inferring states within a finite neighborhood of ground-truth values can be rendered arbitrarily small, even while adversaries retain access to identical communication data streams. This extends classical initial-state privacy to the entire operational timeline under time-varying directed topologies. Numerical examples including an application of cooperative adaptive cruise control demonstrate our proposed algorithm's efficacy.

Index Terms—Platoon control, multi-agent systems, distributed cooperative tracking control, privacy-preserving.

Received 10 October 2023; revised 11 May 2024, 24 September 2024, 8 June 2025, and 14 October 2025; accepted 6 November 2025. Date of publication 10 December 2025; date of current version 18 February 2026. This work was supported by the National Natural Science Foundation of China under Grant 62503114. The work of Peihu Duan was supported by the National Natural Science Foundation of China under Grant 62573054. The work of Rong Su and Yun Lu was supported by the National Research Foundation Singapore under its AI Singapore Programme under Grant AISG4-GC-2023-007-1B. The Associate Editor for this article was S. Santini. (Corresponding author: Lingying Huang.)

Lingying Huang is with the School of Automation, Southeast University, Nanjing 210096, China (e-mail: lingyinghuang@seu.edu.cn).

Rong Su and Yun Lu are with the School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore 639798 (e-mail: RSu@ntu.edu.sg; yun.lu@ntu.edu.sg).

Maode Ma is with the College of Engineering, Qatar University, Doha, Qatar (e-mail: mamaode@qu.edu.qa).

Peihu Duan is with the State Key Laboratory of Environment Characteristics and Effects for Near-Space, Beijing Institute of Technology, Beijing 100081, China (e-mail: duanpeihu@bit.edu.cn).

Digital Object Identifier 10.1109/TITS.2025.3639277

I. INTRODUCTION

THE development of cooperative-tracking technologies, particularly platoon control, has emerged as a pivotal research direction in intelligent transportation systems (ITS) due to its potential to enhance road capacity, fuel efficiency, and traffic safety [1], [2], [3]. In platoon control systems, vehicles maintain tight inter-vehicle spacing through coordinated acceleration/deceleration, requiring real-time exchange of motion states (e.g., position and velocity) among neighboring vehicles. However, existing platoon control strategies often overlook the critical issue of privacy preservation in vehicle-to-vehicle (V2V) communications. Sensitive vehicle data, such as real-time trajectories, could be exploited by malicious actors to infer driving patterns or even identities, posing significant risks to user privacy [4].

Recent advances in distributed control frameworks [5], [6], [7], [8], [9] have enabled resilient platoon coordination without centralized infrastructure. Yet, these methods typically assume full-state sharing among vehicles, leaving privacy vulnerabilities unaddressed. Existing privacy-preserving consensus approaches fall into four categories: anonymity, cryptography, perturbation and state decomposition. While anonymity-based methods protect participant identities [10], they often require centralized infrastructure [11]. Cryptography methods [12], [13] offers strong privacy guarantees but incur high computational and communicational cost. While differential privacy (DP) mechanisms have been explored for generic multi-agent systems [14] on consensus problem, their direct application to platoon control faces two key challenges: (1) DP-induced noise may trades off accuracy for privacy [15], [16], [17]. State decomposition addresses this tradeoff by splitting agent states into visible and hidden substates [18], later extended to directed graphs [19]. (2) The static state nature of consensus problems contrasts sharply with the dynamic state evolution in cooperative-tracking control. Limited work addresses privacy in the latter context. Xu et al. [20] pioneered a state decomposition-based method for leader-follower systems but imposed two restrictive assumptions: (i) stable state matrices (often invalid in tracking scenarios) and (ii) communication signals independent of node states (compromising string stability and failing to bound coupling errors). These limitations motivate our work on a generalized privacy-preserving constrained cooperative-tracking control framework

applicable to unstable dynamics while explicitly bounding coupling errors.

Our solution navigates the dual challenge of coupling error boundedness and privacy preservation, where strict error bounding risks state exposure. The framework accommodates time-varying heterogeneous directed topologies and transmission delays, relaxing conventional B-strong connectivity requirements to infinite-flow networks—a fundamental consensus condition. We consider both internal and external adversaries [19]. The main contributions are listed below:

- 1) **Generalized Communication Topology:** Departing from B-strongly-connected digraphs [21], we establish consensus guarantees under the infinite-flow property, a necessary condition for achieving cooperative tracking consensus, for time-varying heterogeneous directed networks.
- 2) **Offline PPCC Control:** A push-sum based privacy-preserving constrained cooperative-tracking (PPCC) control algorithm (**Algorithm 1**) ensures the bounded coupling errors with privacy guarantees. Crucially, the error bound depends solely on the initial state, enabling offline control gain design (**Theorem 1**). We further characterize the asymptotic convergence factor's relationship to control gains (**Theorem 2**).
- 3) **Trajectory privacy** We prove that our proposed method can preserve privacy against internal and external adversaries (**Theorem 3, 4**). Our analysis demonstrates that the probability of an adversary inferring states within a finite neighborhood of true values can be rendered arbitrarily small, even while adversaries leverage all the communication data streams. This formulation strengthens conventional initial-state privacy preservation by extending protection to the entire operational timeline, thereby establishing **trajectory privacy** for dynamic systems.

The paper is organized as follows: Section II outlines mathematical preliminaries. Section III formalizes the problem. Section IV details the PPCC algorithm and stability analysis. Section V provides privacy proofs. Section VI presents case studies, and Section VII concludes.

II. MATHEMATICAL PRELIMINARIES

A. Notations

$\mathbb{N}$ is the set of natural numbers. $\mathbb{R}$ and $\mathbb{R}^n$ represent the set of real numbers and n -dimensional column vectors, respectively. $(\cdot)^\top$, $\|\cdot\|$ represent the transpose and 2-norm of $(\cdot)$, respectively. $U[a, b]$ represents a uniform distribution between $[a, b]$, where $a \leq b$. For the matrices $A \in \mathbb{R}^{n \times m}$ and $B \in \mathbb{R}^{p \times q}$, $A \otimes B \in \mathbb{R}^{np \times mq}$ represents their Kronecker product. Let $\rho(A)$ represent the spectral radius of matrix A . The column vector with all entries equal to 1 is denoted by $\mathbf{1}$ and the identity matrix is I , and their sizes are determined from the context. Let $\{M(t)\}$ be a sequence of matrices, $M(k:s) = M(k)M(k-1) \cdots M(s)$, $k \geq s \in \mathbb{N}$, where $M(s:s) = M(s)$.

B. Graph Theory

We model a communication network as a directed graph (digraph) $\mathcal{G} = (\mathcal{N}, \mathcal{E})$, where $\mathcal{N} = \{1, 2, \dots, N\}$ is the set of

nodes and $\mathcal{E} \subseteq \mathcal{N} \times \mathcal{N}$ is the edge set consisting of ordered pairs of nodes. We use $(j, i) \in \mathcal{E}$ to represent that a communication link exists from node j to node i . Given a nonnegative matrix $M = [m_{ij}] \in \mathbb{R}^{N \times N}$, a directed graph induced by M is denoted by $\mathcal{G}_M = (\mathcal{N}, \mathcal{E}_M)$, where $(j, i) \in \mathcal{E}_M$ if and only if $m_{ij} > 0$. Since node i can always access its own information, we set $m_{ii} \geq 0$, where $m_{ii} = 0$ means that node i chooses to ignore information about itself, and $(i, i) \in \mathcal{E}_M$ if $m_{ii} > 0$. Given $\mathcal{G}_M$, the in-neighbour set and out-neighbour set of node i are respectively defined as

$$\begin{aligned} \mathcal{N}_{M,i}^{\text{in}} &= \{j \in \mathcal{N} : (j, i) \in \mathcal{E}_M\}, \\ \mathcal{N}_{M,i}^{\text{out}} &= \{j \in \mathcal{N} : (i, j) \in \mathcal{E}_M\}. \end{aligned} \quad (1)$$

A digraph is strongly connected if there exists at least one directed path from any node i to any node j .

III. SYSTEM MODEL AND PROBLEM FORMULATION

A. Communication Topology and Agent Dynamics

We consider a network consisting N agents, represented by the set $\mathcal{N}$. The communication between agents is represented by a time-varying digraph topology $\mathcal{G}_{M_k}$ captured by $M(k) = [m_{ij}(k)]$ at each time step $k \in \mathbb{N}$. In this setup, each agent can dynamically select neighbors for message transmission. While the out-degree of each agent is known (as it controls its outgoing links), the in-degree remains unknown due to the stochastic nature of neighbor selection by other agents.

Definition 1 (Strong Aperiodicity): A sequence $\{M(k)\}$ is strongly aperiodic if there exists $\eta > 0$ such that $m_{ii}(k) \geq \eta$, for all $t \geq 0$ and $i \in \mathcal{N}$.

Definition 2 (DIFP): The sequence $\{M(k)\}$ has the directed infinite flow property (DIFP) if for any nontrivial $S \subset \mathcal{N}$, $\sum_{t=0}^{\infty} m_{ij}(t) = \infty$, $\forall i \in S, j \in \bar{S} \triangleq \mathcal{N} \setminus S$.

According to [22, Theorem 3.1], having the DIFP is the crucial condition for guaranteeing consensus in a deterministic network topology. We assume that $\{M(k)\}$ is strongly aperiodic and has the DIFP. The strong aperiodicity implies that each agent's state at time k directly influences its state at $k+1$, while the DIFP guarantees finite-time reachability between any pair of agents [23].

Assumption 1: The graph $\mathcal{G}_{M_k}$ is generated from $\{M(k)\}$ which is strongly aperiodic and has the DIFP.

Let $k_0 = 0$, and for any $r \geq 1$, define the time instants k_r recursively as

$$k_r = \arg \min_{t' > k_{r-1}} \left\{ \left(\min_{S \subset \mathcal{N}} \sum_{t=k_{r-1}}^{t'-1} m_{S\bar{S}}(t) \right) > 0 \right\}, \quad (2)$$

where $m_{S\bar{S}}(t) = \sum_{i \in S, j \in \bar{S}} m_{ij}(t)$. This ensures that $M(k_r - 1 : k_{r-1})$ is strongly connected for all $r \geq 1$, and k_r is finite due to the DIFP. Define $l_0 = 0$ and the time intervals

$$l_r = k_r - k_{(r-1)N}. \quad (3)$$

Let $x_i(k) \in \mathbb{R}^n$ denotes the state of node (agent) i at time $k\tau$. The dynamics of agent i are governed by:

$$x_i(k+1) = Ax_i(k) + Bu_i(k), \forall i \in \mathcal{N}, \forall k \in \mathbb{N}, \quad (4)$$

where $u_i(k) \in \mathbb{R}^m$ represents the control signal of agent i , and $A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times m}$ are the state and input matrices, respectively.

Assumption 2: The pair (A, B) is stabilizable.

B. Problem Formulation

We address a constrained cooperative-tracking problem, where each agent i aims to asymptotically track a reference trajectory $p_0(k) \in \mathbb{R}^n$ governed by

$$p_0(k+1) = Ap_0(k), \forall k \in \mathbb{N}. \quad (5)$$

Define the coupling error between agents i and j as $z_{ij}(k) \triangleq x_i(k) - x_j(k)$.

Definition 3: The control reaches constrained cooperative-tracking consensus if:

- 1) $\lim_{k \rightarrow \infty} \|x_i(k) - p_0(k)\| = 0, \forall i \in \mathcal{N}$;
- 2) There exists a constant $D \leq 0$ such that $\|z_{ij}(k)\| \leq D, \forall i \neq j \in \mathcal{N}$ and $\forall k \in \mathbb{N}$.

A common control strategy is given by

$$u_i(k) = K \left[\sum_{(j,i) \in \mathcal{E}_{M_k}} m_{ij}(k)(x_j(k) - x_i(k)) + h_i(k)(p_0(k) - x_i(k)) \right], \quad (6)$$

where $K \in \mathbb{R}^{m \times n}$ is a constant control gain, and $h_i(k) > 0$ denotes the link weight from the leader to agent i (zero otherwise). Prior works [24], [25] have established the necessity of parameter K to achieve cooperative-tracking consensus. The coupling error in this protocol allows adaptive tuning of K to ensure boundedness.

Given the DIFP of $\mathcal{G}_{M_k}$, all agents will eventually receive the leader's state $p_0(k)$ after $k \geq l_1 + 1$ time steps. However, privacy constraints limit the transmission of $p_0(k)$, making the worst-case analysis critical for privacy-preserving designs.

Remark 1: Constrained cooperative-tracking is essential in applications like formation control, where agents must maintain a predefined formation vector $H = [H_1^T, H_2^T, \dots, H_N^T]^T \in \mathbb{R}^{Nn}$ while adhering to safety constraints. For example, consider agents with states $\tilde{x}_i(k) = [\tilde{x}_{p,i}(k), \tilde{x}_{v,i}(k)]^T \in \mathbb{R}^2$ (position and velocity). The dynamics matrix becomes $A = \begin{bmatrix} 1 & \tau \\ 0 & 1 \end{bmatrix}$. The formation is reached iff. $\lim_{k \rightarrow \infty} \|\tilde{x}_i(k) - H_i - p_0(k)\| = 0, \forall i \in \mathcal{N}$. If $A(H_i - H_j) = H_i - H_j$, implying $H_{v,i} = H_{v,j}$, where $H_{v,i}$ denotes the velocity difference of agent i with respect to (w.r.t.) the leading node, signifying a collective aim for uniform velocity in formation control, the formation consensus tracking can be transformed into the above cooperative tracking framework by letting $x_i(k) = \tilde{x}_i(k) - H_i$ and satisfying the first condition of Definition 3. To avoid collisions, the safety constraint $|\tilde{x}_i(k) - \tilde{x}_j(k)| \geq (|i - j| - 1)d_0 + l_0$ must hold, where l_0 is the minimum safety gap. This requirement is satisfied by setting $D = d_0 - l_0$ in Definition 3.

Direct transmission of nodes' state information among neighbouring nodes poses privacy risks in the presence of adversaries. The main goal of our paper is to design a privacy-preserving constrained cooperative-tracking control. The main challenge lies in ensuring the boundedness of the coupling error while preventing leakage of the agents' actual states.

IV. PUSH-SUM BASED PRIVACY-PRESERVING CONSTRAINED COOPERATIVE-TRACKING CONTROL

To address the privacy challenges in cooperative tracking control, we propose a novel integration of the push-sum algorithm [26] with state decomposition techniques [18]. Our approach decomposes each agent's state into two complementary substates, with only one substate being shared through neighborhood communications. The push-sum framework ensures bounded coupling error through constrained evolution of both decomposed states and their averaged consensus.

A. General Push-Sum Algorithm

The push-sum consensus protocol operates through dual state-weight propagation. For each agent $i \in \mathcal{N}$, we maintain:

- $y_i(k)$: Cumulative state estimate,
- $w_i(k)$: Associated weight parameter.

The average state estimate is the ratio $\bar{x}_i(k) = \frac{y_i(k)}{w_i(k)}$ at each iteration k . The detail of the updates follows: the coupling weight from node i to j at time step k is denoted as $q_{ji}(k)$, where $q_{ji}(k) = \frac{m_{ji}(k)}{\sum_{j \in \mathcal{N}_{M_k,i}^{out}} m_{ji}(k)} \forall i, j \in \mathcal{N}$. At each time step k , node i splits y_i and w_i into shares $S_{i \rightarrow j}(k) = (q_{ji}(k)y_i(k), q_{ji}(k)w_i(k))$ and sends to its out-neighbours $j \in \mathcal{N}_{M_k,i}^{out}$. Then after receiving $q_{ir}(k)y_r(k)$ and $q_{ir}(k)w_r(k)$ from its in-neighbours $r \in \mathcal{N}_{M_k,i}^{in}$, node i updates y_i and w_i follows

$$\begin{cases} y_i(k+1) = \sum_{r \in \mathcal{N}_{M_k,i}^{in}} q_{ir}(k)y_r(k), \\ w_i(k+1) = \sum_{r \in \mathcal{N}_{M_k,i}^{in}} q_{ir}(k)w_r(k). \end{cases} \quad (7)$$

Finally, the node i updates state $x_i(k) = \frac{y_i(k)}{w_i(k)}$.

Under Assumption 1 and following [23], the algorithm guarantees $\lim_{k \rightarrow \infty} \left\| x_i(k) - \frac{1}{N} \sum_{i=1}^N x_i(0) \right\| = 0$.

B. Push-Sum Based Privacy-Preserving Algorithm Description

Traditional push-sum algorithms face privacy vulnerabilities [27], where an honest-but-curious adversary can reconstruct initial states by eavesdropping on node communications. This vulnerability stems from direct transmission of weighted state information. To address this limitation, we propose a novel state decomposition scheme with hidden states that preserves privacy while maintaining convergence properties.

Our solution extends beyond the initial value constraints discussed in [20] by introducing a bounded coupling error mechanism that satisfies Definition 3. The proposed algorithm achieves dual objectives: (1) driving $x_i(k)$ towards the average $\bar{x}(k) \triangleq \frac{\sum_i x_i(k)}{N}$, and (2) ensuring this average tracks a prescribed trajectory $p_0(k)$. The state decomposition mechanism enables distributed average estimation without exposing individual node states.

The proposed push-sum based privacy-preserving constrained cooperative-tracking (PPCC) control in Algorithm 1 introduces virtual nodes i' for each physical node $i \in \mathcal{N}$. This dual-node architecture maintains:

- Visible states $(y_i^{(1)}, w_i^{(1)})$ for physical node i ,
- Hidden states $(y_i^{(2)}, w_i^{(2)})$ for physical node i' .

Algorithm 1 PPCC Control

Initialization: For each node $\forall i \in \mathcal{N}$, randomly initializes visible states $y_i^{(1)}(0) \sim U[-H, H]$ and $w_i^{(1)}(0) \sim U[0, 2]$, where H can be specified by the user based on capabilities of the digital platform. Initialize the hidden states $y_i^{(2)}(0) = 2x_i(0) - y_i^{(1)}(0)$ and $w_i^{(2)}(0) = 2 - w_i^{(1)}(0)$. The intermediate variables $\bar{y}_i^{(l)}$ for $l = 1, 2$, are used to store the updated $y_i^{(l)}$ after consensus step before the dynamic evolution.

State Update: For iteration $k \geq 0$, each node $i \in \mathcal{N}$

1: Randomly chooses $0 < \alpha_i(k) < 1$.

2: Normalizes the coupling weight

$$q_{ji}(k) = \frac{m_{ji}(k)}{\alpha_i(k) + \sum_{j \in \mathcal{N}_{M_{k,i}}^{\text{out}}} m_{ji}(k)}, \forall j \in \mathcal{N}. \quad (8)$$

3: Computes $q_{ji}(k)y_i^{(1)}(k)$ and $q_{ji}(k)w_i^{(1)}(k)$ and send them to its out-neighbours $j \in \mathcal{N}_{M_{k,i}}^{\text{out}}$.

4: **Consensus:** Physical node i

$$\begin{cases} \bar{y}_i^{(1)}(k) = \sum_r q_{ir}(k)y_r^{(1)}(k) + y_i^{(2)}(k), \\ w_i^{(1)}(k+1) = \sum_r q_{ir}(k)w_r^{(1)}(k) + w_i^{(2)}(k). \end{cases} \quad (9)$$

Virtual node i'

$$\begin{cases} \bar{y}_i^{(2)}(k) = \alpha_i(k)y_i^{(1)}(k), \\ w_i^{(2)}(k+1) = \alpha_i(k)w_i^{(1)}(k). \end{cases} \quad (10)$$

5: **Tracking:** Drives $y_i^{(l)}(k)$ to track $w_i^{(l)}(k+1)p_0(k)$, for $l = 1, 2$, i.e.,

$$\begin{aligned} y_i^{(l)}(k+1) &= A\bar{y}_i^{(l)}(k) \\ &+ BKA(w_i^{(l)}(k+1)p_0(k) - \bar{y}_i^{(l)}(k)). \end{aligned} \quad (11)$$

6: **Control:** Generates the control signal as follows

$$u_i(k) = K \left(\frac{y_i^{(2)}(k+1)}{w_i^{(2)}(k+1)} - Ax_i(k) \right). \quad (12)$$

7: Updates its state according to (4).

Key algorithmic phases:

- **Consensus:** achieves agreement between physical and virtual nodes
- **Tracking:** aligns system average with reference trajectory
- **Control:** generates privacy-preserving control signals

C. Convergence Analysis

Let the augmented state vector be defined as $y(k) \triangleq [y_1^{(1)}(k)^\top, \dots, y_N^{(1)}(k)^\top, y_1^{(2)}(k)^\top, \dots, y_N^{(2)}(k)^\top]^\top$, with corresponding definitions $\bar{y}(k)$ and $w(k)$. Then dynamics in Algorithm 1 admit the compact matrix representation:

$$\begin{cases} \bar{y}(k) = \tilde{Q}(k) \otimes I_y(k), \\ y(k+1) = \tilde{Q}(k) \otimes (A - BKA)y(k) \\ + W(k+1) \otimes BKA p(k), \\ w(k+1) = \tilde{Q}(k)w(k), \end{cases} \quad (13)$$

where $\tilde{Q}(k) = \begin{bmatrix} Q(k) & I \\ \Lambda(k) & 0 \end{bmatrix} = [\tilde{q}_{ij}(k)]$, with $Q(k) = [q_{ij}(k)]$, $\Lambda(k) = \text{diag}(\alpha_1(k), \dots, \alpha_N(k))$, $W(k) = \text{diag}(w(k))$, and

$p(k) = [p_0(k)^\top, \dots, p_0(k)^\top]^\top \in \mathbb{R}^{2Nn}$ being the reference trajectory. This leads to the expanded state evolution:

$$\begin{aligned} y(k+1) &= \prod_{t=0}^k \tilde{Q}(t) \otimes (A - BKA)^{k+1} y(0) \\ &+ W(k+1) \otimes \sum_{t=0}^k (A - BKA)^{k-t} BKA p(t). \end{aligned} \quad (14)$$

Define the cumulative interaction matrix $\tilde{M}(k) = \prod_{t=0}^k \tilde{Q}(t) = [\tilde{m}_{ij}(k)] \in \mathbb{R}^{2N \times 2N}$, and its infinite-horizon counterpart $\tilde{M} = \prod_{t=0}^{\infty} \tilde{Q}(t)$. The column stochasticity of $\tilde{M}(k)$ follows inductively from the column-stochastic nature of each $\tilde{Q}(t)$. Let the minimum interaction strength be

$$\epsilon = \min_{i \in \mathcal{N}} \{\alpha_i(k), \min_{j \in \mathcal{N}_{M_{k,i}}^{\text{in}}} q_{ij}(k)\}. \quad (15)$$

Lemma 1 establishes the critical convergence property.

Lemma 1: Under Assumption 1 and $\sum_{r=1}^{\infty} \epsilon^{l_r} = \infty$, the infinite product $\tilde{M}$ converges to a rank-one matrix $v\mathbf{1}^\top$, where $v = [v_i]$ is a stochastic vector.

Proof: Following [28], convergence requires:

$$\lim_{k \rightarrow \infty} \tilde{m}_{ij}(k) - \tilde{m}_{il}(k) = 0, \forall i, j, l \in \{1, 2, \dots, 2N\}. \quad (16)$$

Under Assumption 1, we have that $Q(k_r - 1 : k_{r-1})$ is strongly connected. From the definition of $\tilde{Q}(k)$, any node i' can reach node i with one time step, $\forall i \in \mathcal{N}$, and vice versa. Let $l'_r = l_0 = 0$ and $l'_r = k_{r2N} - k_{(r-1)2N} = l_{2r} + l_{2r-1}$ with $r \geq 1$. Consider the graph generated by $\tilde{M}(r) = \tilde{Q}(l'_r - 1 : l'_r - 1) = [\tilde{m}_{ij}(r)] \in \mathbb{R}^{2N \times 2N}$. Since in length l'_r , there exists a link between any node i to any node j , for all $i, j \in \{1, 2, \dots, 2N\}$; then we have $\tilde{m}_{ij}(r) > 0$. In addition, since $\tilde{Q}(k)$ is column stochastic, one has $\tilde{m}_{ij}(r) > \epsilon^{l'_r}$. Since

$$\begin{aligned} |\tilde{m}_{ij}(k) - \tilde{m}_{il}(k)| &\leq \max_j \tilde{m}_{ij}(k) - \min_j \tilde{m}_{ij}(k) \\ &\leq \prod_{r=1}^{r^*} (1 - \epsilon^{l'_r}) \leq \exp \left(- \sum_{r=1}^{r^*} \epsilon^{l'_r} \right) \end{aligned}$$

using the fact that $\ln(\zeta) \leq \zeta - 1$ for all $\zeta > 0$, where $r^* = \arg \max_r (k_{rN} \leq k)$, then (16) is equivalent to prove that $\lim_{r^* \rightarrow \infty} \sum_{r=1}^{r^*} \epsilon^{l'_r} = \infty$. We can derive that

$$\sum_{r=1}^{r^*} \epsilon^{l'_r} \geq \epsilon^{\min_r l_r} \max \left\{ \sum_{r=1}^{r^*} \epsilon^{l_{2r}}, \sum_{r=1}^{r^*} \epsilon^{l_{2r-1}} \right\}.$$

From $\sum_{r=1}^{\infty} \epsilon^{l_r} = \infty$ and $\epsilon > 0$, one has $\lim_{r^* \rightarrow \infty} \max \left\{ \sum_{r=1}^{r^*} \epsilon^{l_{2r}}, \sum_{r=1}^{r^*} \epsilon^{l_{2r-1}} \right\} = \infty$; therefore, $\lim_{r^* \rightarrow \infty} \sum_{r=1}^{r^*} \epsilon^{l'_r} = \infty$, which completes the proof. ■

Note that l_r calculated by (3), pertains to the original communication topology before state-decomposition, whereas $\tilde{M}$ is defined for the state-decomposition system. Lemma 1 ensures the convergence of $\tilde{M}$ by choosing $\{\alpha_i(k)\}$ to satisfy (15) within the original network.

Remark 2: $\mathcal{G}_{M_k}$ strongly connected at each time step k is a special case which satisfies $\tilde{M} = v\mathbf{1}^\top$ since $\lim_{r^* \rightarrow \infty} r^* \epsilon = \infty$.

Let $\hat{\bar{x}}(0)$ denote the initial estimate of the system average, where $\hat{\bar{x}}(0) = \bar{x}(0)$. The dynamic evolution of $\hat{\bar{x}}(k)$ follows

$$\hat{\bar{x}}(k+1) = A\hat{\bar{x}}(k) + BKA(p_0(k) - \hat{\bar{x}}(k)). \quad (17)$$

Define the estimation error $\hat{e}_i(k) \triangleq x_i(k) - \hat{\bar{x}}(k)$. Since $\|z_{ij}(k)\| \leq \|\hat{e}_i(k)\| + \|\hat{e}_j(k)\|$, the second requirement of Definition 3 is satisfied if $\|\hat{e}_i(k)\| \leq \frac{D}{2}$. Let the true (estimated) tracking errors be defined as $e_i(k) \triangleq x_i(k) - p_0(k)$ ($e_p(k) \triangleq \hat{\bar{x}}(k) - p_0(k)$).

Theorem 1: Under Assumptions 1 and 2, the constrained cooperative-tracking consensus can be reached by Algorithm 1 if the following conditions hold:

- 1) $\|\hat{e}_i(0)\| \leq \frac{D}{2}$;
- 2) The gain matrix K ensures $\gamma = \rho(A - BKA) < 1$ and $\gamma^k \|\hat{e}_i(0)\| + \gamma^k k \|BK\| \left(\|e_p(0)\| + \max_{1 \leq i \leq N, l=1,2} \left\| \frac{y_i^{(l)}(1)}{w_i^{(l)}(1)} - \hat{\bar{x}}(1) \right\| \right) \leq \frac{D}{2}, \forall k$;
- 3) the sequence of the column-stochastic matrices $\{\tilde{Q}(k)\}$ satisfies $\sum_{r=1}^{\infty} \epsilon^r = \infty$ where l_r is defined in (3).

Proof: Since $\|e_i(k)\| \leq \|\hat{e}_i(k)\| + \|e_p(k)\|$, the convergence of the tracking error is equivalent to ensure the convergence of $\|e_p(k)\|$ and $\|\hat{e}_i(k)\|$. By (12), one has

$$\begin{aligned} e_i(k+1) &= (A - BKA)e_i(k) \\ &+ BK \left(\frac{y_i^{(2)}(k+1)}{w_i^{(2)}(k+1)} - Ap_0(k) \right). \end{aligned} \quad (18)$$

To analyze $\frac{y_i^{(2)}(k+1)}{w_i^{(2)}(k+1)}$, we apply Lemma 1 and (14)

$$\begin{aligned} &\lim_{k \rightarrow \infty} \frac{y_i^{(2)}(k+1)}{w_i^{(2)}(k+1)} \\ &= \lim_{k \rightarrow \infty} \frac{1}{[\tilde{M}w(0)]_{N+i}} \left[\tilde{M} \otimes (A - BKA)^{k+1} y(0) \right. \\ &\quad \left. + W(k+1) \otimes \sum_{t=0}^k (A - BKA)^{k-t} BKA p(t) \right]_{N+i} \\ &= \lim_{k \rightarrow \infty} \frac{v_{i+N}}{v_{i+N} \mathbf{1}^\top w(0)} \left((A - BKA)^{k+1} \mathbf{1}^\top y(0) \right. \\ &\quad \left. + \mathbf{1}^\top w(0) \sum_{t=0}^k (A - BKA)^{k-t} BKA p_0(k) \right) \\ &= \lim_{k \rightarrow \infty} \left((A - BKA)^{k+1} \bar{x}(0) + \sum_{t=0}^k (A - BKA)^{k-t} BKA p(t) \right). \end{aligned} \quad (19)$$

From (17), one has $\hat{\bar{x}}(k+1) = (A - BKA)^{k+1} \bar{x}(0) + \sum_{t=0}^k (A - BKA)^{k-t} BKA p(t)$; then along with (19), one has $\lim_{k \rightarrow \infty} \frac{y_i^{(2)}(k+1)}{w_i^{(2)}(k+1)} = \hat{\bar{x}}(k+1)$. Similarly, we can also derive $\lim_{k \rightarrow \infty} \left\| \frac{y_i^{(1)}(k+1)}{w_i^{(1)}(k+1)} - \hat{\bar{x}}(k+1) \right\| = 0$. Define

$$\delta_k = \max_{1 \leq i \leq N, l=1,2} \left\| \frac{y_i^{(l)}(k+1)}{w_i^{(l)}(k+1)} - \hat{\bar{x}}(k+1) \right\|. \quad (20)$$

From the above analysis, $\lim_{k \rightarrow \infty} \delta_k = 0$. Substituting into (18) yields

$$\|e_i(k+1)\| \leq \gamma \|e_i(k)\| + \|BK\|(\delta_k + \|e_p(k+1)\|). \quad (21)$$

Given $\gamma = \rho(A - BKA) < 1$, one has $\|e_p(k+1)\| \leq \gamma^{k+1} \|e_p(0)\|$ by (17). Substituting into (21), we have

$$\begin{aligned} \|e_i(k+1)\| &\leq \gamma^{k+1} (\|e_i(0)\| + (k+1) \|BK\| \|e_p(0)\|) \\ &+ \|BK\| \sum_{t=0}^k \gamma^{k-t} \delta_t. \end{aligned} \quad (22)$$

Since $\lim_{k \rightarrow \infty} \delta_k = 0$ and $0 < \gamma < 1$, we conclude $\lim_{k \rightarrow \infty} \sum_{t=0}^{k-1} \gamma^{k-1-t} \delta_t = 0$ [29, Lemma 3.1(a)]. Therefore, the tracking goal is achieved, i.e., $\lim_{k \rightarrow \infty} \|e_i(k)\| = 0, \forall i \in \mathbb{N}$.

Next, we study the requirement to ensure the constrained coupling error. From (4), (12), (17), the error update for agent i is derived as

$$\begin{aligned} \hat{e}_i(k+1) &= (A - BKA)\hat{e}_i(k) \\ &+ BK \frac{y_i^{(2)}(k+1) - w_i^{(2)}(k+1) Ap_0(k)}{w_i^{(2)}(k+1)}. \end{aligned} \quad (23)$$

Using the triangle inequality and the definition of δ_k (20), the coupling term is bounded as

$$\begin{aligned} &\|y_i^{(2)}(k+1) - w_i^{(2)}(k+1) Ap_0(k)\| \\ &\leq \|y_i^{(2)}(k+1) - w_i^{(2)}(k+1) \hat{\bar{x}}(k+1)\| \\ &\quad + w_i^{(2)}(k+1) \|\hat{\bar{x}}(k+1) - p_0(k+1)\| \\ &\leq w_i^{(2)}(k+1) (\delta_k + \gamma^{k+1} \|e_p(0)\|). \end{aligned} \quad (24)$$

This step leverages the decomposition of estimation errors and the exponential decay of γ^k . Substituting the bound into the error dynamics yields

$$\begin{aligned} \|\hat{e}_i(k+1)\| &\leq \gamma \|\hat{e}_i(k)\| + \|BK\|(\delta_k + \gamma^{k+1} \|e_p(0)\|) \\ &\leq \gamma^{k+1} (\|\hat{e}_i(0)\| + (k+1) \|BK\| \|e_p(0)\|) + \|BK\| \sum_{t=0}^k \gamma^{k-t} \delta_t. \end{aligned}$$

This shows the error decays exponentially with a residual term dependent on initial conditions. Define $\zeta(k) = W(k)^{-1} \otimes I y(k)$. Substituting into the system dynamics reveals

$$\begin{aligned} \zeta(k+1) &= W(k+1)^{-1} \tilde{Q}(k) \otimes (A - BKA) y(k) + I \otimes BKA p(k) \\ &= W(k+1)^{-1} \tilde{Q}(k) W(k) \otimes (A - BKA) \zeta(k) \\ &\quad + I \otimes BKA p(k). \end{aligned} \quad (25)$$

The average consensus term $\bar{\zeta}(k) = \frac{1}{2N} \mathbf{1}^\top \otimes I \zeta(k)$ satisfies

$$\bar{\zeta}(k+1) = (A - BKA) \bar{\zeta}(k) + BKA p_0(k). \quad (26)$$

Since $\bar{\zeta}(0) = \hat{\bar{x}}(0)$, we have $\bar{\zeta}(k) = \hat{\bar{x}}(k)$. Define the consensus error $e_\zeta(k) = \zeta(k) - \mathbf{1} \otimes \bar{\zeta}(k)$. Subtracting (26) from the individual agent dynamics (25) gives

$$e_\zeta(k+1) = W(k+1)^{-1} \tilde{Q}(k) W(k) \otimes (A - BKA) e_\zeta(k). \quad (27)$$

This leads to the per-agent bound

$$e_{\zeta,i}(k+1) = (A - BKA) \frac{\sum_{j=1}^{2N} \tilde{q}_{ij}(k) w_j(k) e_{\zeta,j}(k)}{\sum_{j=1}^{2N} \tilde{q}_{ij}(k) w_j(k)}, \quad (28)$$

which implies that

$$\|e_{\zeta,i}(k+1)\| \leq \gamma \max_j \|e_{\zeta,j}(k)\|, \quad (29)$$

since for any $\{a_k\}_{1 \leq k \leq K} \geq 0$, $\{b_k\}_{1 \leq k \leq K} > 0$, $\frac{\sum_{k=1}^K a_k}{\sum_{k=1}^K b_k} \leq \max_k \frac{a_k}{b_k}$. By (20) and (29), one has $\delta_k = \max_j \|e_{\zeta,j}(k+1)\| \leq \gamma^k \delta_0$. Combining the bounds, the total error satisfies $\|\hat{e}_i(k)\| \leq \gamma^k \|\hat{e}_i(0)\| + \gamma^k k \|BK\| (\|e_p(0)\| + \delta_0)$. Under conditions 1) and 2), this ensures $\|\hat{e}_i(k)\| \leq \frac{D}{2}$, $\forall k \in \mathbb{N}$, guaranteeing constrained consensus. ■

Remark 3: From the proof part, define

$$\bar{e}_i(k) \triangleq \gamma^k \bar{e}_i(0) + \gamma^k k \|BK\| (\|e_p(0)\| + \delta_0), \forall k \in \mathbb{N}. \quad (30)$$

with $\bar{e}_i(0) = \max\{\|\hat{e}_i(0)\|, \|e_i(0)\|\}$. One has that $\|e_i(k)\|$ and $\|\hat{e}_i(k)\|$ are upper bounded by $\|\bar{e}_i(k)\|$, which means that the control under Algorithm 1 will drive each node to track $p_0(k)$ while minimizing the coupling error asymptotically.

Remark 4: From (30), the error bound depends only on initial conditions, allowing the feedback gain K to be designed offline. This is a critical advantage for real-time implementation. In addition, if K satisfies $\gamma = \rho(A - BKA) < 1$ and

$$\|BK\| \leq \frac{(1-\gamma)\|\hat{e}_i(0)\|}{\|e_p(0)\| + \delta_0} \leq \frac{(1-\gamma)D}{2(\|e_p(0)\| + \delta_0)}, \quad (31)$$

one further has $\|\hat{e}_i(k+1)\| \leq \|\hat{e}_i(k)\| \leq \dots \leq \|\hat{e}_i(0)\| \leq \frac{D}{2}$. Specifically, For systems with $\rho(A) \leq 1$, such K always exists if $\|BK\|$ is sufficiently small.

When the communication graph $\mathcal{G}M_k$ is strongly connected, the infinite sum $\sum r = 1^\infty \epsilon^r = \infty$ ensures convergence even with relaxed topology requirements, i.e., 3) in Theorem 1.

D. Asymptotic Convergence Factor

Define the composite error vector $\bar{e}(k) \triangleq [\bar{e}_1(k), \dots, \bar{e}_N(k)]^\top$ by concatenating the individual error vectors $\bar{e}_i(k)$. Under Algorithm 1, the asymptotic convergence factor ϱ_{asym} is formally defined as

$$\varrho_{\text{asym}} \triangleq \sup_{\bar{e}(0) \neq 0} \lim_{k \rightarrow \infty} \left(\frac{\|\bar{e}(k)\|}{\|\bar{e}(0)\|} \right)^{\frac{1}{k}}, \quad (32)$$

which is characterized by the following theorem.

Theorem 2: Under the conditions specified in Theorem 1, the asymptotic convergence factor of Algorithm 1 satisfies:

$$\varrho_{\text{asym}} = \gamma, \quad (33)$$

where $\gamma = \rho(A - BKA)$.

Proof: From the recursive relation (30), we derive the lower bound:

$$\begin{aligned} \varrho_{\text{asym}} &= \sup_{\bar{e}(0) \neq 0} \lim_{k \rightarrow \infty} \left(\frac{\|\bar{e}(k)\|}{\|\bar{e}(0)\|} \right)^{\frac{1}{k}} \\ &\geq \lim_{k \rightarrow \infty} \gamma \left(1 + Nk^2 \|BK\|^2 \frac{\|e_p(0)\|^2}{\|\bar{e}(0)\|^2} \right)^{\frac{1}{2k}} = \gamma. \end{aligned} \quad (34)$$

For the upper bound, recall from Theorem 1 that $\delta_k = \max_j \|e_{\zeta,j}(k+1)\|$. By the same argument, we obtain:

$$\sup_{\delta_0 \neq 0} \lim_{k \rightarrow \infty} \left(\frac{\delta_k}{\delta_0} \right)^{\frac{1}{k}} \leq \gamma.$$

Combining this with (34) establishes the equality $\varrho_{\text{asym}} = \gamma$, completing the proof. ■

Remark 5: Under Assumption 2, minimizing the asymptotic convergence factor ϱ_{asym} requires selecting the feedback gain K to minimize $\gamma = \rho(A - BKA)$.

Remark 6: Since $\bar{e}_i(k)$ serves as an upper bound for the tracking error $\|e_i(k)\|$, the asymptotic convergence rate of $\|e_i(k)\|$ is strictly less than or equal to ϱ_{asym} . Furthermore, since $\|z_{ij}(k)\| \leq \|\hat{e}_i(k)\| + \|\hat{e}_j(k)\| \leq 2\bar{e}_i(k)$, its asymptotic convergence factor is also bounded by ϱ_{asym} .

V. PRIVACY-PRESERVING PERFORMANCE ANALYSIS

We analyze the privacy guarantees of the proposed algorithm under two adversarial models [19]:

- (1) **Internal adversaries:** Honest-but-curious nodes ($\mathcal{A}$) that collude to infer the states of non-adversarial nodes ($\mathcal{L} = \mathcal{N} \setminus \mathcal{A}$) using the information set $\mathcal{I}_{\mathcal{A}} = \bigcup_{k=0}^{\infty} (\bigcup_{a \in \mathcal{A}} \mathcal{I}_a(k))$, where $\mathcal{I}_a(k)$ contains

$$\begin{aligned} \mathcal{I}_a(k) &\triangleq \{q_{ja}(k), \alpha_a(k), x_a(k), q_{ab}(k)y_b^{(1)}(k), \\ &q_{ab}(k)w_b^{(1)}(k), y_a^{(l)}(k), w_a^{(l)}(k) | j \in \mathcal{N}, b \in \mathcal{N}, l = 1, 2\}. \end{aligned} \quad (35)$$

- (2) **External adversaries:** Eavesdroppers with full network topology knowledge and access to all transmitted data, using the information set $\mathcal{I}_e = \bigcup_{k=0}^{\infty} \mathcal{I}_e(k)$, where

$$\begin{aligned} \mathcal{I}_e(k) &\triangleq \\ &\{q_{ij}(k), q_{ij}(k)y_j^{(1)}(k), q_{ij}(k)w_j^{(1)}(k) | \forall i \neq j \in \mathcal{N}\}. \end{aligned} \quad (36)$$

Let $\mathcal{I}_{\mathcal{A}}$ be any feasible information set collected all the trajectory's information for the considered adversary, regardless of temporal pooling. Define $\Delta(\mathcal{I}_{\mathcal{A}}, i, k)$ as the set of all possible states $x_i(k)$ consistent with $\mathcal{I}_{\mathcal{A}}$ under Algorithm 1. The privacy preservation is formalized as follows.

Definition 4: Node i 's privacy is preserved if for any finite $\varepsilon > 0$ and $k \in \mathbb{N}$,

$$\mathbb{E}[1_{\varepsilon}(\Delta(\mathcal{I}_{\mathcal{A}}, i, k))] = 0,$$

where $1_{\varepsilon}(\cdot)$ is the indicator function:

$$1_{\varepsilon}(S) = \begin{cases} 1 & \text{if } \exists x_i(k)', x_i(k)'' \in S \text{ s.t. } \|x_i(k)' - x_i(k)''\| \leq \varepsilon, \\ 0 & \text{otherwise.} \end{cases}$$

The subscript k in Definition 4 denotes the specific time step under inference consideration, rather than restricting the adversary's temporal data access. Since the information set $\mathcal{I}_{\mathcal{A}}$ contains all historical data up to time k , we evaluate the worst-case privacy leakage. The probabilistic guarantee ensures that for any finite time horizon k , the likelihood of an adversary successfully inferring states within an ε -neighborhood of actual values can be made arbitrarily small. This formulation strengthens conventional initial-state privacy preservation by extending protection to the entire operational timeline, thereby establishing **trajectory privacy** for dynamic systems. The asymptotic behavior reveals that as k tends to infinity, all follower nodes asymptotically converge to the leader's state under cooperative-tracking consensus. This convergence property implies that diameter analysis remains meaningful only for finite time horizons $k \in \mathbb{N}$. Crucially, adversaries cannot determine the temporal threshold for consensus achievement,

restricting them to inference attempts based on the available information set $\mathcal{I}_A$, where $\mathcal{I}_A = \mathcal{I}_A$ for the internal adversaries and $\mathcal{I}_A = \mathcal{I}_e$. The subsequent theorems formalize these privacy preservation guarantees.

Theorem 3: The privacy of any node $i \in \mathcal{L}$ can be preserved against the set of honest-but-curious nodes $\mathcal{A}$ with information set $\mathcal{I}_A$ under Algorithm 1.

Proof: We prove this theorem through an enhanced analysis considering $\mathcal{A} = \mathcal{N} \setminus \{i\}$. For any $x_i(k) \in \Delta(\mathcal{I}_A, i, k)$ where $k \in \mathbb{N}$, let $x_i(k)' = x_i(k) + e(k)$ with arbitrary $e(k)$ elements. We demonstrate the existence of modified values

$$\{y_i^{(l)}(0)', w_i^{(l)}(0)', q_{ji}(k)', \alpha_i(k)' \mid j \in \mathcal{N}, l = 1, 2\}$$

such that $x_i(k)' \in \Delta(\mathcal{I}_A, i, k)$.

Consider $k = 0$. Let $x_i(0)' = x_i(0) + \frac{c}{2}$ and

$$\begin{aligned} y_i^{(1)}(0)' &= y_i^{(1)}(0) + d_1, y_i^{(2)}(0)' = y_i^{(2)}(0) + c - d_1, \\ w_i^{(1)}(0)' &= w_i^{(1)}(0) + d_2, w_i^{(2)}(0)' = w_i^{(2)}(0) - d_2, \end{aligned} \quad (37)$$

where $c, d_1 \in \mathbb{R}^n$ are vectors and $d_2 \in \mathbb{R}$ is a scalar to make $x_i(0)' \in \Delta(\mathcal{I}_A, i, 0)$. We choose

$$\begin{aligned} d_1 &= \frac{d_2}{w_i^{(1)}(0)} y_i^{(1)}(0), \\ q_{ji}(0)' &= \frac{w_i^{(1)}(0)}{w_i^{(1)}(0) + d_2} q_{ji}(0), \forall j \neq i \in \mathcal{N}. \end{aligned} \quad (38)$$

Maintaining column stochasticity requires

$$\alpha_i(0)' + q_{ii}(0)' = 1 - \frac{w_i^{(1)}(0)}{w_i^{(1)}(0) + d_2} (1 - \alpha_i(0) - q_{ii}(0)).$$

Selecting parameters

$$d_2 = w_i^{(1)}(0), q_{ii}(0)' = \frac{1 + q_{ii}(0)}{2}, \alpha_i(0)' = \frac{\alpha_i(0)}{2}, \quad (39)$$

preserves key quantities: $\bar{y}_i^{(1)}(0)' = \bar{y}_i^{(1)}(0)$, $w_i^{(1)}(1)' = w_i^{(1)}(1)$ and $\bar{y}_i^{(2)}(0)' = \bar{y}_i^{(2)}(0)$, $w_i^{(2)}(1)' = w_i^{(2)}(1)$ from (10). Subsequent evolution via (11) yields $y_i^{(1)}(1)' = y_i^{(1)}(1)$, $y_i^{(2)}(1)' = y_i^{(2)}(1)$. Thus, for $\bigcup_{a \in \mathcal{A}} \mathcal{I}_a(k)$ with $k \in \mathbb{N}^+$, maintaining $\bar{Q}(k)' = \bar{Q}(k)$ preserves the information set $\mathcal{I}_A$ accessible to $\mathcal{N} \setminus \{i\}$, which means $x_i(0)' \in \Delta(\mathcal{I}_A, i, 0)$.

Next, considering the inferred state step $k \in \mathbb{N}^+$ with the parameterization in (37)-(39) and $\bar{Q}(k)' = \bar{Q}(k)$, $\forall k \in \mathbb{N}^+$, we obtain $x_i(k)' = x_i(k) + (A - BKA)^k \frac{c}{2}$. The privacy indicator function becomes:

$$\mathbf{1}_\varepsilon(\Delta(\mathcal{I}_A, i, k)) = \begin{cases} 1 & \text{if } \|(A - BKA)^k \frac{c}{2}\| \leq \varepsilon, \\ 0 & \text{otherwise.} \end{cases}$$

Note that the range of c can be restricted by the range of the initial state $x_i(0)$. However, this information is not available to the internal adversary, they may make a guess of the element of c to be any real number. With the elements of c uniformly distributed in $\mathbb{R}$, we obtain

$$\mathbb{E}[\mathbf{1}_\varepsilon(\Delta(\mathcal{I}_A, i, k))] = 0. \quad \blacksquare$$

Remark 7: Unlike [18], [19], our approach does not require that there exists at least one legitimate neighbour to compensate the difference caused by the deviation of the initial

position since honest-but-curious neighbors cannot distinguish tracking errors from consensus errors.

Next, we consider the external adversaries with as much information as possible excluding the weights and states of hidden nodes. Ensuring privacy against such external adversaries guarantees the preservation of each node's state against other external adversaries with lesser information.

Theorem 4: The privacy of any node i is preserved against external adversaries with information set $\mathcal{I}_e$ under Algorithm 1.

Proof: Since external adversaries know $q_{ji}(k)$ for $j \neq i$, we maintain indistinguishability by $y_i^{(1)}(k)' = y_i^{(1)}(k)$ and $w_i^{(1)}(k)' = w_i^{(1)}(k)$. For any $x_i(k) \in \Delta(\mathcal{I}_e, i, k)$, we construct

$$\{y_i^{(2)}(0)', w_i^{(1)}(0)', w_i^{(2)}(0)', q_{ii}(k)', \alpha_i(k)'\}$$

such that $x_i(k)' = x_i(k) + e(k) \in \Delta(\mathcal{I}_e, i, k)$.

For $k = 0$, let $x_i(0)' = x_i(0) + \frac{d}{2} y_i^{(1)}(0)$ with

$$\begin{aligned} y_i^{(2)}(0)' &= y_i^{(2)}(0) + d y_i^{(1)}(0), q_{ii}(0)' = d + q_{ii}(0), \\ w_i^{(1)}(0)' &= w_i^{(1)}(0) = 0, w_i^{(2)}(0)' = w_i^{(2)}(0) = 2, \end{aligned} \quad (40)$$

where scalar d satisfies $\epsilon \leq d + q_{ii}(0) \leq 1 - \epsilon$ for strong aperiodicity. This preserves $\bar{y}_i^{(1)}(0)' = \bar{y}_i^{(1)}(0)$, leading to $y_i^{(1)}(1)' = y_i^{(1)}(1)$ and $w_i^{(1)}(1)' = w_i^{(1)}(1)$ via (11). Considering $\mathcal{I}_e(k)$ with $k \in \mathbb{N}^+$, maintaining $\bar{Q}(k)' = \bar{Q}(k)$ preserves the information set $\mathcal{I}_e$, which means $x_i(0)' \in \Delta(\mathcal{I}_e, i, 0)$.

For $k \in \mathbb{N}^+$, choosing parameterization in (40) and maintaining $\bar{Q}(k)' = \bar{Q}(k)$ yields

$$x_i(k)' = x_i(k) + \frac{d}{2} (A - BKA)^k y_i^{(1)}(0).$$

The privacy measure becomes

$$\mathbf{1}_\varepsilon(\Delta(\mathcal{I}_A, i, k)) = \begin{cases} 1 & \text{if } \frac{d}{2} \|(A - BKA)^k y_i^{(1)}(0)\| \leq \varepsilon, \\ 0 & \text{otherwise.} \end{cases}$$

Since the range of initialization H is unknown to the external adversary, they may make a guess of the elements of $y_i^{(1)}(0)$ to be any real number. With $y_i^{(1)}(0)$ elements perceived as uniformly distributed in $\mathbb{R}$, we obtain

$$\mathbb{E}[\mathbf{1}_\varepsilon(\Delta(\mathcal{I}_A, i, k))] = 0. \quad \blacksquare$$

Remark 8: Regarding method itself, H is set by the user based on the capabilities of the digital platform. While H must be large enough to ensure the expectation in Definition 4 is small, ε should reflect the desired privacy granularity. In practice, ε can be tuned based on the application's sensitivity, e.g., choosing $\varepsilon = 0.1$ for fine-grained privacy versus choosing $\varepsilon = 10$ for coarser guarantees. Additionally, adversaries' lack of knowledge about H creates effective infinite uncertainty regarding initialization ranges. In this case, we can interpret $H = \infty$, which satisfies the privacy guarantee in Definition 4.

VI. NUMERICAL SIMULATION

In this section, we first compare with the existing privacy-preserving control in a stable dynamic example. Next, an application of cooperative adaptive cruise control (CACC) illustrates the effectiveness in an unstable dynamic scenario.

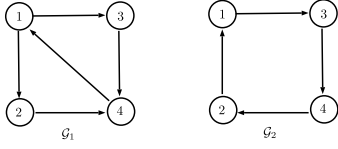
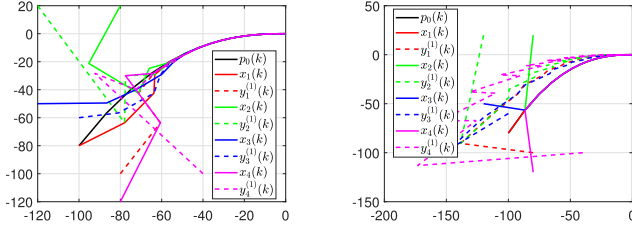
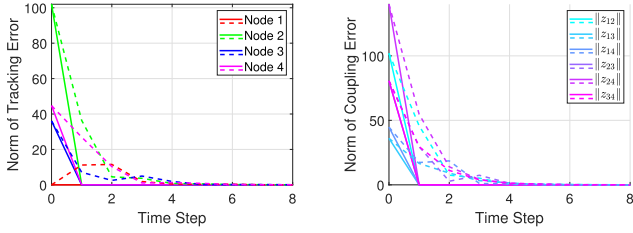


Fig. 1. Demonstration of switching communication topology.



(a) State trajectories under the algorithm [20]. (b) State trajectories under Algorithm 1.



(c) Tracking error. (d) Coupling error.

Fig. 2. Comparison on performances.

A. Stable Dynamic Example

We assume the dynamic of nodes are the same as shown in [20, Example 2], where the state matrix is $A = \begin{bmatrix} \sin(\frac{\pi}{3}) & 0 \\ 0 & \sin(\frac{\pi}{4}) \end{bmatrix}$ and the initial state of the tracking trajectory is $p_0(0) = [-100, -100]^T$. Besides, we consider a four-node network, where $x_1(0) = [-100, -80]^T$, $x_2(0) = [-80, -20]^T$, $x_3(0) = [-120, -50]^T$, $x_4(0) = [-80, -120]^T$.

Consider a switching digraph shown as $\mathcal{G}_1, \mathcal{G}_2$ in Fig. 1. Let the communication topology be $\mathcal{G}_1$ when k is even, and $\mathcal{G}_2$ otherwise. Each vehicle has the same effect of their neighbours. By letting $\alpha_1 = 0.1, \alpha_2 = 0.4, \alpha_3 = 0.2, \alpha_4 = 0.4$,

$$Q(k) = \begin{cases} \begin{bmatrix} 0.3 & 0 & 0 & 0.3 \\ 0.3 & 0.3 & 0 & 0 \\ 0.3 & 0 & 0.4 & 0 \\ 0 & 0.3 & 0.4 & 0.3 \end{bmatrix}, & \text{if } k \text{ is even,} \\ \begin{bmatrix} 0.45 & 0.3 & 0 & 0 \\ 0 & 0.3 & 0 & 0.3 \\ 0.45 & 0 & 0.4 & 0 \\ 0 & 0 & 0.4 & 0.3 \end{bmatrix}, & \text{if } k \text{ is odd.} \end{cases} \quad (41)$$

Let $D = \max_{i,j \in \mathcal{N}} z_{ij}(0)$ and $BK = I$. One has $\gamma = \rho(A - BKA) = 0$ and it satisfies the conditions in Theorem 1. Initialize $w(0) = \mathbf{1}$ and set the transmitted values to be $y_1^{(1)}(0) = [-80, -100]^T$, $y_2^{(1)}(0) = [-120, 20]^T$, $y_3^{(1)}(0) = [-100, -60]^T$, $y_4^{(1)}(0) = [-40, -100]^T$.

Performance comparisons are shown in Fig. 2, where the horizontal and vertical coordinates represent the first and second components of the states. In Fig. 2(a) and (b), solid lines

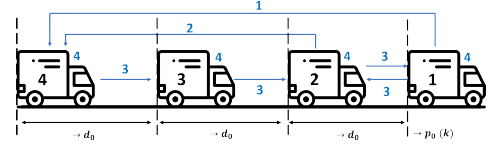


Fig. 3. Demonstration of the platoon communication topology; the blue link represents a communication between vehicles while the number above the link represents the weight of the current link (We assign the weight of the self-loop link as 4).

depict the actual trajectories of nodes, while dashed lines of the same color represent the transmitted trajectories. The privacy of real-time states is effectively preserved, as the transmitted trajectories deviate from the actual ones under the proposed algorithms. Further comparisons of tracking and coupling errors are shown in Figs. 2(c) and (d). Solid lines correspond to results under Algorithm 1, while dashed lines represent the method in [20]. Notably, with $BK = I$, our algorithm achieves convergence of both tracking and coupling errors in a single time step, outperforming the prior approach. This aligns with the theoretical analysis in Section IV-D, where the asymptotic convergence factor is 0. However, the coupling error performance in [20] is suboptimal due to the mismatch between initial states of virtual and actual nodes. This is evident in the crossovers between nodes 1 and 4 in Fig. 2(a). Such issues are mitigated in our algorithm.

B. Unstable Dynamic Example-CACC

Let there be N vehicles in a platoon, with the i -th vehicle indexed as node i in the set $\mathcal{N}$. The communication graph is generated by $M_k = [m_{ij}(k)]$, where the vehicle i separates its own states based on the weight $m_{ij}(k)$. The vehicle's position, velocity, and acceleration are combined to form an augmented state $\tilde{x}_i(k)$, and the states evolve as described in [30]:

$$\dot{\tilde{x}}_i(t) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & -\frac{1}{\tau_e} \end{bmatrix} \tilde{x}_i(t) + \begin{bmatrix} 0 \\ 0 \\ \frac{1}{\tau_e} \end{bmatrix} u_i(t), \quad \forall i \in \mathcal{N}, \quad (42)$$

where τ_e is a time constant related to engine, and u_i is the control to be designed. Assume that the control signal within the time period $[k\tau, (k+1)\tau)$ is constant, denoted by $u_i(k)$. By forward difference approximations, (42) is rewritten as

$$\tilde{x}_i(k+1) = A\tilde{x}_i(k) + Bu_i(k), \quad (43)$$

where $A = \begin{bmatrix} 1 & \tau & \tau^2/2 \\ 0 & 1 & \tau \\ 0 & 0 & 1 - \frac{\tau}{\tau_e} \end{bmatrix}$ and $B = \begin{bmatrix} 0 \\ 0 \\ \frac{\tau}{\tau_e} \end{bmatrix}$. Each vehicle i aims

to track a predefined trajectory $p_0(k)$ following (5), with constant speed and zero acceleration, while maintaining a constant spacing $(i-1)d_0$ by communicating with their neighbouring vehicles, as shown in Fig. 3. Let $x_i(k) = \tilde{x}_i(k) + (i-1)d_0$, and let the safe gap between two successive vehicles be the length of the vehicle l_0 .

As illustrated in Fig. 3, the communication topology can be mathematically expressed through the weight matrix

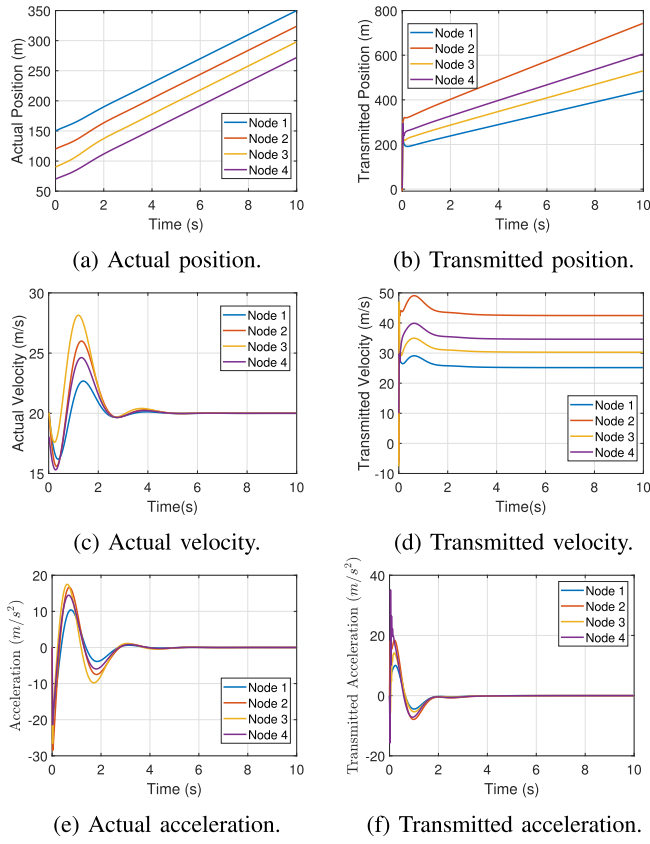


Fig. 4. Demonstration of actual states versus transmitted states under PPCC control given time-invariant digraphs.

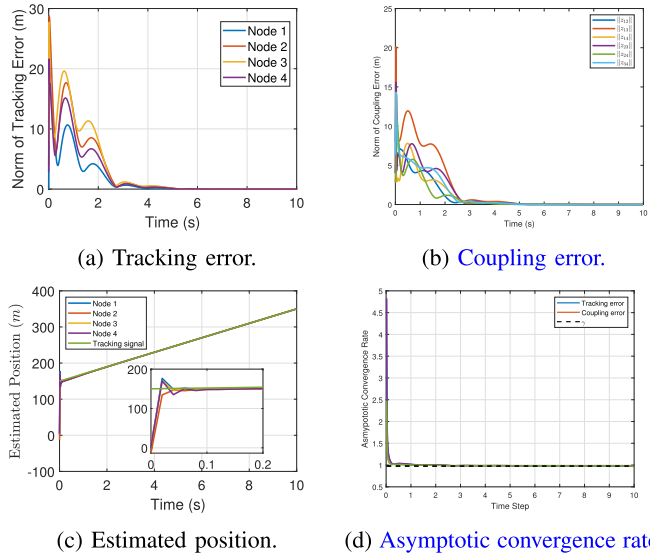


Fig. 5. Demonstration of the convergence and estimation performance under PPCC control given time-invariant digraphs.

$$M = \begin{bmatrix} 4 & 3 & 0 & 0 \\ 3 & 4 & 3 & 0 \\ 0 & 0 & 4 & 3 \\ 1 & 2 & 0 & 4 \end{bmatrix}.$$
 We demonstrate that PPCC control can protect privacy in both time-(in)variant digraphs applied to CACC. We set the engine constant to be $\tau_e = 0.5$ [31] and use a

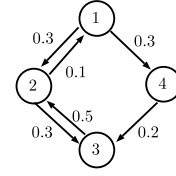


Fig. 6. Coupling weights between vehicles.

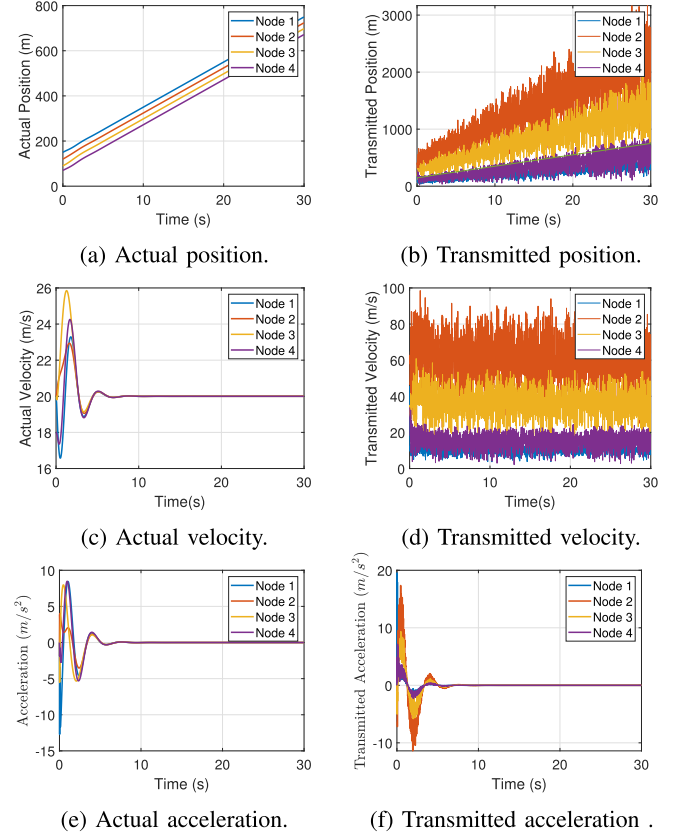


Fig. 7. Demonstration of actual states versus transmitted states under PPCC control given time-variant digraphs.

sampling time $\tau = 0.02s$. Let the predefined trajectory known by the participating vehicles be $p_0(0) = [150, 20, 0]^T$, the initial states of each vehicle in the platoon be $x_1(0) = [150, 20, 0]^T$, $x_2(0) = [120, 20, 0]^T$, $x_3(0) = [90, 20, 0]^T$, $x_4(0) = [70, 18, 0]^T$, the spacing gap be $d_0 = 26m$, and the vehicle length be $l_0 = 6m$. Thus, to avoid collision, the coupling constrained error is set to $D = 20m$. We use the communication graph shown in Fig. 3. Set $H = 10$, we have that $\delta_0 \leq 10$. By choosing $K = [4, 4.8, 0.7]^T$, the enhanced condition (29) is satisfied, which further ensures the constrained cooperative-tracking consensus from Theorem 1.

1) Time-Invariant Communication Digraph: Let $\alpha_1 = 0.1$, $\alpha_2 = 0.2$, $\alpha_3 = 0.2$, $\alpha_4 = 0.3$, where α_i is only known by vehicle i . The actual states $\tilde{x}_i(k)$ and transmitted states $y_i^{(1)}(k)$ are shown in Fig. 4. Furthermore, the formation consensus is reached, as shown in Fig. 5(a). The coupling errors, meeting the constrained requirement, are shown in Fig. 5(b). With $p_{ij}y_j^{(1)}(k)$ and $p_{ij}w_j^{(1)}(k)$ as available information, the estimated

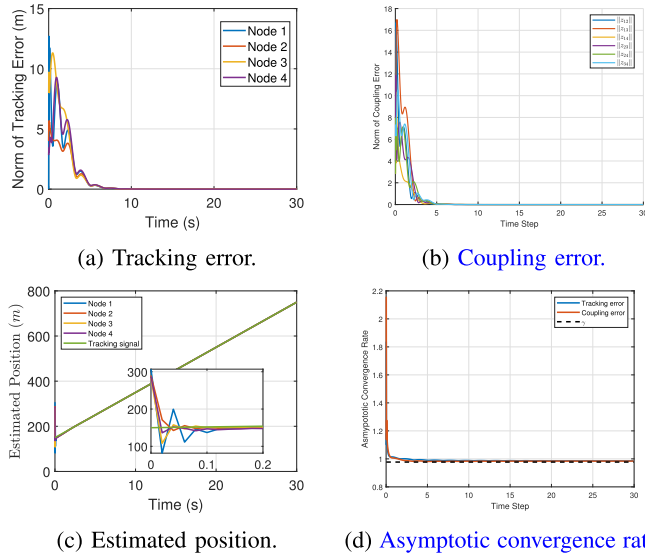


Fig. 8. Demonstration of the convergence and estimation performance under PPCC control given time-variant digraphs.

position obtained by $y_i^{(1)}(k)$, converges to the predefined signal as $k \rightarrow \infty$ in Fig. 5(c). However, especially in the initial few seconds, it differs a lot from the actual position trajectory of each vehicle. From these figures, while transmitted states may asymptotically reflect system trends, adversaries cannot estimate the exact state at any finite k . Lastly, Fig. 5(d) shows the asymptotic convergence rate, corroborating with Theorem 2.

2) *Time-Varying Communication Digraphs*: Consider a special time-varying communication digraph where $q_{ij}(k)$ is fixed, $\forall i, j \in \mathcal{N}, j \neq i$. At each time step k , the vehicle will choose the hidden weight $a_i(k) \in (0, 1)$, and set $q_{ii}(k) = 1 - \sum_j q_{ij}(k) - a_i(k)$. The communication topology, with weights between vehicles, is illustrated in Fig. 6. Set $w(0) = 1$. The actual states $\tilde{x}_i(k)$ and transmitted states $y_i^{(1)}(k)$ are shown in Fig. 7. The tracking error and estimated states shown in Fig. 8 yield results consistent with those observed in time-invariant graphs. From Fig. 7, the transmitted states do not exhibit stable performance and differ a lot from the actual position. Due to this property, if we can transmit $w_i^{(1)}(t)$ in a more secure way that is hard to be intercepted by the external adversaries, those adversaries—despite having knowledge of the coupling weights—will not be able to accurately estimate the actual states. For the internal adversaries, although the exchange of $w_i^{(1)}(t)$ is necessary to update the weights, the actual states is preserved since the estimated position varies from the actual position shown in Fig. 8(c) and Fig. 7(a).

VII. CONCLUSION

In this paper, we introduce a constrained cooperative-tracking consensus control protocol based on state decomposition, specifically tailored for privacy-preserving platoon control in intelligent transportation systems. Our work addresses two critical challenges in vehicular platooning: (1) maintaining bounded coupling errors under dynamic communication topologies, and (2) safeguarding vehicle

state privacy against adversarial threats. The proposed PPCC algorithm advances platoon control methodologies by integrating a push-sum protocol for dynamic consensus tracking with a novel mechanism that exploits hidden-actual state discrepancies to generate control signals. We derive a pivotal condition for the control strategy to achieve constrained cooperative-tracking consensus and analyse the convergence rate, exploring its dependence on the chosen control gain. Additionally, we study privacy of the PPCC control in the presence of both internal and external adversaries. Notably, our results reveal the probability of inferred states being concentrated around the true state can be arbitrarily small while the information available to adversaries remains unchanged, highlighting the effectiveness of our control protocol in safeguarding the privacy of agents' states. We validate our approach through its application in CACC with collision avoidance considerations. Future work includes the design of privacy-preserving cooperative-tracking control strategies tailored to diverse agent dynamics. Moreover, we aim to explore scenarios where agents' access to the leaders' states is restricted, adversaries have knowledge of initial stage interpretation, and control actions carry associated costs.

REFERENCES

- [1] J. Chen, H. Liang, J. Li, and Z. Lv, "Connected automated vehicle platoon control with input saturation and variable time headway strategy," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 8, pp. 4929–4940, Aug. 2021.
- [2] Z. Gao, X. Li, Z. Wei, W. Liu, G. Guo, and S. Wen, "Observer-based secure predefined-time control of vehicular platoon systems under attacks in sensors and actuators," *IEEE Trans. Intell. Transp. Syst.*, vol. 26, no. 6, pp. 8907–8918, Jun. 2025.
- [3] Y. Gong and W.-X. Zhu, "Robust control for the mixed vehicular platoon with polytopic uncertainties under non-ideal communication environment: A gain-scheduling control-based approach," *IEEE Trans. Veh. Technol.*, vol. 74, no. 9, pp. 13682–13698, Sep. 2025.
- [4] G. Cormode, C. Procopiuc, D. Srivastava, E. Shen, and T. Yu, "Differentially private spatial decompositions," in *Proc. IEEE 28th Int. Conf. Data Eng.*, Apr. 2012, pp. 20–31.
- [5] W. Liu and J. Huang, "Leader-following consensus for linear multiagent systems via asynchronous sampled-data control," *IEEE Trans. Autom. Control*, vol. 65, no. 7, pp. 3215–3222, Jul. 2020.
- [6] Y. Liu, D. Yao, H. Li, and R. Lu, "Distributed cooperative compound tracking control for a platoon of vehicles with adaptive NN," *IEEE Trans. Cybern.*, vol. 52, no. 7, pp. 7039–7048, Jul. 2022.
- [7] Y.-S. Ma, W.-W. Che, C. Deng, and Z.-G. Wu, "Data-driven distributed vehicle platoon control for heterogeneous nonlinear vehicle systems," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 3, pp. 2373–2381, Mar. 2024.
- [8] M. Zhang, C. Wang, W. Zhao, J. Liu, and Z. Zhang, "A multi-vehicle self-organized cooperative control strategy for platoon formation in connected environment," *IEEE Trans. Intell. Transp. Syst.*, vol. 26, no. 3, pp. 4002–4018, Mar. 2025.
- [9] S. Huang, C. Zhao, L. Huang, P. Cheng, J. Wu, and L. Cai, "Plug-and-play distributed estimation of driving states in an open vehicle platoon," *IEEE Trans. Ind. Informat.*, vol. 20, no. 4, pp. 5328–5338, Apr. 2024.
- [10] B. C. Fung, K. Wang, A. W.-C. Fu, and S. Y. Philip, *Introduction to Privacy-preserving Data Publishing: Concepts and Techniques*. Boca Raton, FL, USA: CRC Press, 2010.
- [11] B. Gedik and L. Liu, "Protecting location privacy with personalized k-anonymity: Architecture and algorithms," *IEEE Trans. Mobile Comput.*, vol. 7, no. 1, pp. 1–18, Jan. 2008.
- [12] Y. Lu and M. Zhu, "Privacy preserving distributed optimization using homomorphic encryption," *Automatica*, vol. 96, pp. 314–325, Oct. 2018.
- [13] W. Ding, J. Zhou, W. Yang, and Y. Tang, "An efficient encoding mechanism against eavesdropper with side channel information," *Automatica*, vol. 153, Jul. 2023, Art. no. 111062.

- [14] C. Dwork, "Differential privacy," in *International Colloquium on Automata, Languages, and Programming* (Lecture Notes in Computer Science), vol. 4052. Berlin, Germany: Springer, 2006, pp. 1–12.
- [15] Z. Huang, S. Mitra, and N. Vaidya, "Differentially private distributed optimization," in *Proc. 16th Int. Conf. Distrib. Comput. Netw.*, Jan. 2015, pp. 1–10.
- [16] Y. Wang and A. Nedić, "Tailoring gradient methods for differentially private distributed optimization," *IEEE Trans. Autom. Control*, vol. 69, no. 2, pp. 872–887, Feb. 2024.
- [17] L. Huang, J. Wu, D. Shi, S. Dey, and L. Shi, "Differential privacy in distributed optimization with gradient tracking," *IEEE Trans. Autom. Control*, vol. 69, no. 9, pp. 5727–5742, Sep. 2024.
- [18] Y. Wang, "Privacy-preserving average consensus via state decomposition," *IEEE Trans. Autom. Control*, vol. 64, no. 11, pp. 4711–4716, Nov. 2019.
- [19] X. Chen, L. Huang, K. Ding, S. Dey, and L. Shi, "Privacy-preserving push-sum average consensus via state decomposition," *IEEE Trans. Autom. Control*, vol. 68, no. 12, pp. 7974–7981, Dec. 2023.
- [20] H. Xu, Y.-H. Ni, Z. Liu, and Z. Chen, "Privacy-preserving leader-following consensus via node-augment mechanism," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 68, no. 6, pp. 2117–2121, Jun. 2021.
- [21] A. Nedic and A. Olshevsky, "Distributed optimization over time-varying directed graphs," *IEEE Trans. Autom. Control*, vol. 60, no. 3, pp. 601–615, Mar. 2015.
- [22] B. Touri, *Product of Random Stochastic Matrices and Distributed Averaging*. Cham, Switzerland: Springer, 2012.
- [23] P. Rezaeian, B. Ghareisifard, T. Linder, and B. Touri, "Push-sum on random graphs: Almost sure convergence and convergence rate," *IEEE Trans. Autom. Control*, vol. 65, no. 3, pp. 1295–1302, Mar. 2020.
- [24] J. Hu and Y. Hong, "Leader-following coordination of multi-agent systems with coupling time delays," *Phys. A, Stat. Mech. Appl.*, vol. 374, no. 2, pp. 853–863, Feb. 2007.
- [25] K. You and L. Xie, "Network topology and communication data rate for consensusability of discrete-time multi-agent systems," *IEEE Trans. Autom. Control*, vol. 56, no. 10, pp. 2262–2275, Oct. 2011.
- [26] D. Kempe, A. Dobra, and J. Gehrke, "Gossip-based computation of aggregate information," in *Proc. 44th Annu. IEEE Symp. Found. Comput. Sci.*, Aug. 2004, pp. 482–491.
- [27] H. Gao, C. Zhang, M. Ahmad, and Y. Wang, "Privacy-preserving average consensus on directed graphs using push-sum," in *Proc. IEEE Conf. Commun. Netw. Secur. (CNS)*, May 2018, pp. 1–9.
- [28] E. Seneta, *Non-negative Matrices and Markov Chains*. Cham, Switzerland: Springer, 2006.
- [29] S. Sundhar Ram, A. Nedić, and V. V. Veeravalli, "Distributed stochastic subgradient projection algorithms for convex optimization," *J. Optim. Theory Appl.*, vol. 147, no. 3, pp. 516–545, Dec. 2010.
- [30] B. Besselink and K. H. Johansson, "A delay-based spacing policy for vehicle platooning: Analysis and control," in *Time Delay Systems*. Cham, Switzerland: Springer, 2017, pp. 249–262.
- [31] V. L. Bageshwar, W. L. Garrard, and R. Rajamani, "Model predictive control of transitional maneuvers for adaptive cruise control vehicles," *IEEE Trans. Veh. Technol.*, vol. 53, no. 5, pp. 1573–1585, Sep. 2004.



Lingying Huang received the B.S. degree in electrical engineering and automation from Southeast University, Jiangsu, China, in 2017, and the Ph.D. degree in electrical and computer engineering from The Hong Kong University of Science and Technology, Hong Kong, in 2021. She was a Research fellow at the School of Electrical and Electronic Engineering, Nanyang Technological University, from 2021 to 2024. She is currently a Young Chief Professor with Southeast University. Her current research interests include intelligent vehicles, cyber-physical system security/privacy, and distributed optimization.



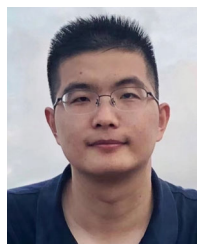
Rong Su (Senior Member, IEEE) received the Bachelor of Engineering degree from the University of Science and Technology of China and the Master of Applied Science and Ph.D. degrees from the University of Toronto. He was affiliated with the University of Waterloo and the Technical University of Eindhoven, before he joined Nanyang Technological University in 2010. His research interests include multi-agent systems, discrete-event system theory, model-based fault diagnosis, cyber security analysis and synthesis, control and optimization of complex networks with applications in flexible manufacturing, intelligent transportation, human–robot interface, power management, and green buildings. In the aforementioned areas, he has more than 240 journals and conference publications, one monograph, and 14 granted/filed patents. He was a recipient of the 2021 Hsue-shen Tsien Paper Award from *IEEE/CAA Journal of Automatica Sinica*, a Distinguished Lecturer of the 2020 Chinese Conference on Decision and Control (CCDC'20), and an IEEE Distinguished Lecturer of the IEEE Robotics and Automation Society. He was the Chair of the Technical Committee on Smart Cities in the IEEE Control Systems Society from 2016 to 2019. He is the Co-Chair of the Technical Committee on Automation in Logistics in the IEEE Robotics and Automation Society and the chair of the Control Systems Chapter, Singapore. He is an Associate Editor for *Automatica* (IFAC), *Journal of Discrete Event Dynamic Systems: Theory and Applications*, and *Journal of Control and Decision*.



Maode Ma (Senior Member, IEEE) received the Ph.D. degree from the Department of Computer Science, The Hong Kong University of Science and Technology, in 1999. He is currently a Research Professor with the College of Engineering, Qatar University, Qatar. He has over 460 international academic publications, including more than 220 journal articles and over 230 conference papers. His publication has received about 7600 citations in Google Scholar. He is a fellow of IET. He is a Senior Member of the IEEE Communication Society and a member of ACM. He is the Chair of ACM and Singapore Chapter. He serves as the Editor-in-Chief for *International Journal of Computer and Communication Engineering* and *Journal of Communications*. He serves as a Senior Editor for *IEEE COMMUNICATIONS SURVEYS AND TUTORIALS* and an Associate Editor for *International Journal of Wireless Communications and Mobile Computing* and *International Journal of Communication Systems*.



Yun Lu (Member, IEEE) received the Ph.D. degree in mechanical engineering from Beijing Institute of Technology, Beijing, China, in 2020. From January 2019 to March 2020, he was a Visiting Scholar with the Department of Mechanical Engineering, University of Michigan, Ann Arbor, MI, USA. He is currently a Research Fellow with the School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore. He is the author of several refereed journal articles in *IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS* and *IEEE TRANSACTIONS ON NEURAL SYSTEMS AND REHABILITATION ENGINEERING*. His research interests include human behavior modeling, platoon control, driver assistance systems, machine learning, and brain-control driving. He was selected as an outstanding graduate in Beijing in 2020.



Peihu Duan received the B.Eng. degree in mechanical engineering from the Huazhong University of Science and Technology, Wuhan, China, in 2015, and the Ph.D. degree in mechanical systems and control from Peking University, Beijing, China, in 2020. From 2019 to 2025, he conducted research as a Research Assistant and a Post-Doctoral Fellow at the City University of Hong Kong, The Hong Kong University of Science and Technology, The University of Hong Kong, and the KTH Royal Institute of Technology. Currently, he is an Assistant Professor with Beijing Institute of Technology, Beijing, China. His research interests include distributed state estimation and control for multi-agent systems and resource scheduling in sensor networks.